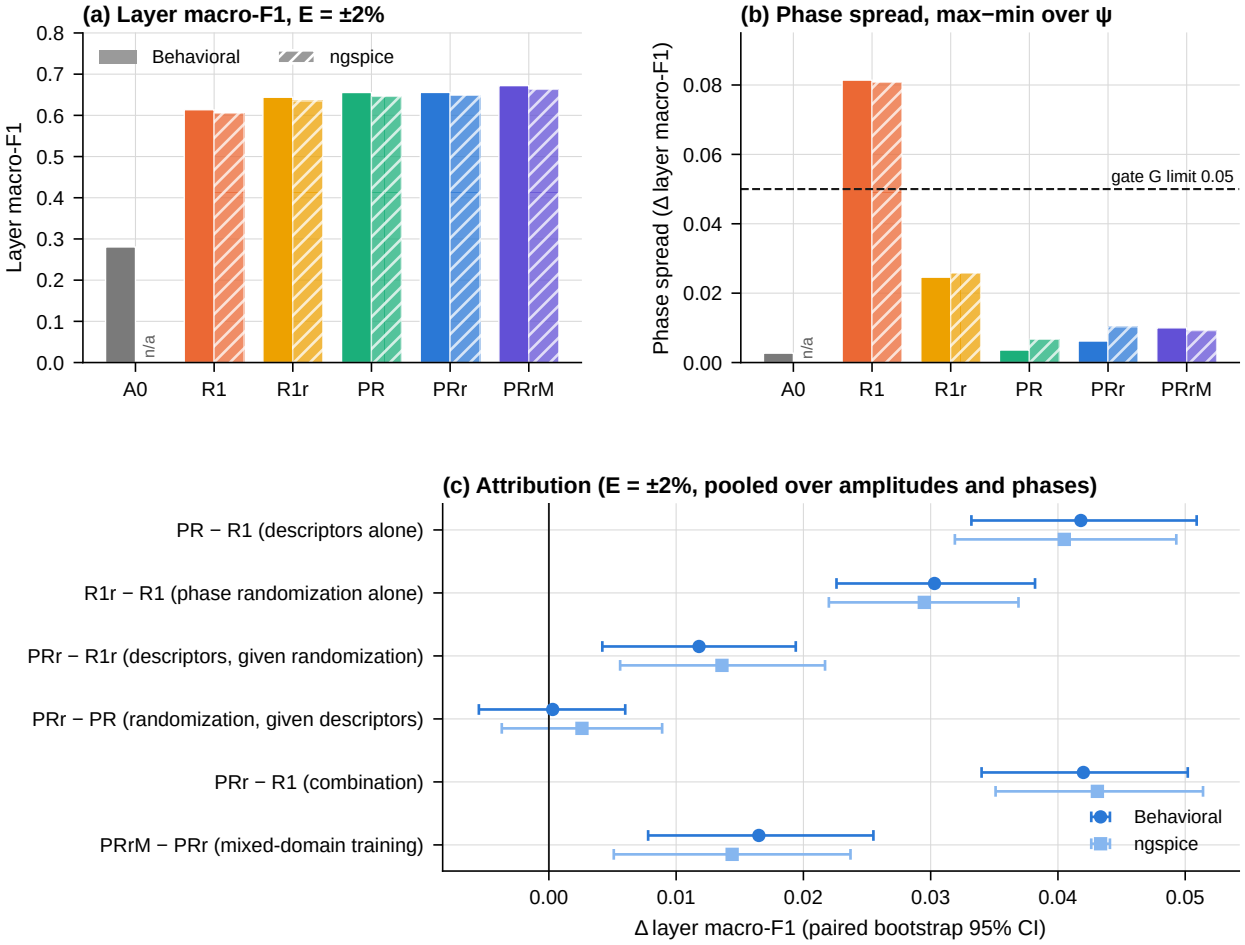
(a) Layer macro-F1, E = ±2%
Behavioral
ngspice
Layer macro-F1
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
n/a
A0
R1
R1r
PR
PRr
PRrM
(b) Phase spread, max−min over ψ
Phase spread (Δ layer macro-F1)
0.08
0.06
0.04
0.02
0.00
gate G limit 0.05
n/a
A0
R1
R1r
PR
PRr
PRrM
(c) Attribution (E = ±2%, pooled over amplitudes and phases)
PR − R1 (descriptors alone)
R1r − R1 (phase randomization alone)
PRr − R1r (descriptors, given randomization)
PRr − PR (randomization, given descriptors)
PRr − R1 (combination)
PRrM − PRr (mixed-domain training)
Behavioral
ngspice
0.00
0.01
0.02
0.03
0.04
0.05
Δ layer macro-F1 (paired bootstrap 95% CI)